

You may work in a small group on this assignment. The goal of this assignment is to give you practice in preparing different datasets for downstream analysis work.

We'll look at some of the most interesting student solutions during our meetup on the following Wednesday.

Data Tidying and Transformation

Working with tidyr and dplyr in R

1. Objective

This project develops practical competency in transforming wide-format datasets into tidy formats suitable for downstream analysis. All transformations are performed using the tidyr and dplyr packages in R.

2. Dataset Selection

You must satisfy all three of the following requirements when selecting your datasets:

1. Select **three (3) distinct wide-format datasets** from the Week 5 Discussion 5A items.
2. You may substitute your own dataset **only if it is structurally wide**.
3. Do **not** use the instructor's Sample Post dataset.

NOTE Each dataset must be independent and analyzed separately. Reusing the same data source across multiple submissions will not satisfy the three-dataset requirement.

3. Data Preparation

Complete all three sub-tasks below for each selected dataset. Steps 3.1 through 3.3 must be applied independently to every dataset.

3.1 Raw Data Construction

Create a source data file that captures all information from the original dataset while preserving its wide (untidy) structure. This file serves as the unmodified input to your transformation pipeline.

- Create a **.csv** file containing **all information** from the original dataset.

- Preserve the original wide format — do not tidy the data at this stage.
- Alternative: store the dataset in a relational database instead of a flat `.csv` file.

DELIVERABLE Commit the raw dataset file(s) to your GitHub repository before proceeding to Step 3.2.

3.2 Data Import and Tidying

Primary Grading Component

Import your data into R and apply a reproducible sequence of `tidyr` and `dplyr` operations. Your code must demonstrate each of the following transformations explicitly:

- Reshape from wide to tidy (long) format
- Normalize variable structure
- Rename variables to follow a consistent naming convention
- Address missing or inconsistent values with documented handling decisions

The following example illustrates a minimal tidying pipeline using `pivot_longer()`:

```
library(tidyr)
library(dplyr)

# Import raw data
raw_data <- read.csv("data/dataset_wide.csv")

# Reshape to tidy format
tidy_data <- raw_data |>
  pivot_longer(
    cols      = -id,
    names_to  = "variable",
    values_to = "value"
  ) |>
  rename_with(tolower) |>
  drop_na()
```

NOTE The majority of your grade is based on the correctness, clarity, and reproducibility of this step. Transformations must be fully executable from a clean R session with no manual intervention.

3.3 Analysis

Perform the specific analysis requested in the original Discussion 5A item for each dataset. All analysis must operate on the tidy dataset produced in Step 3.2 — do not analyze the raw wide-format data directly.

Include the following outputs as appropriate for each analysis:

- Summary tables with clearly labeled columns and units
- Visualizations with axis labels, titles, and legends

- A brief narrative interpreting each result in context

TIP Use `ggplot2` for visualizations. Consistent visual styling across your three datasets improves readability and demonstrates intentional analytical design.

4. Documentation and Reproducibility

All code and narrative must be written in a **Quarto Markdown (.Qmd)** file. Quarto extends the R Markdown ecosystem with improved cross-referencing, theming, and multi-format output — see the quarto.org documentation if you are new to the format.

Your .Qmd file must document each of the following in order:

4. Data source — where the dataset originated and how it was obtained
5. Data structure before tidying — column names, row count, and a representative sample
6. Transformation steps — commented, executable code with explanations of each decision
7. Analytical methods — a description of the statistical or summarization approach applied
8. Conclusions — a substantive interpretation of results with reference to specific values

Your .Qmd file must also meet the following standards:

- All code chunks must execute without error from a clean R session
- Code must be readable — use consistent indentation, spacing, and comments
- Set a global seed (`set.seed()`) if any analysis involves randomness

```
---
title: "Project 2 - Data Tidying"
author: "Your Name"
date: today
format:
  html:
    toc: true
    code-fold: true
    theme: flatly
execute:
  echo: true
  warning: false
---
```

NOTE Quarto (.Qmd) is the successor to R Markdown (.Rmd) and is fully supported by RStudio. If you are working in an environment that does not yet support Quarto, consult your instructor before submission.

5. Submission Requirements

Submit the following two URLs for each of your three datasets. All six links must be functional at the time of grading.

9. URL to the `.Qmd` source file in your GitHub repository
10. URL to the rendered, published output on `rpubs.com`

IMPORTANT All datasets and code must be publicly accessible via GitHub. Private repositories will not be graded. Verify that all links resolve without authentication before submitting.

Submission Checklist

Before submitting, confirm each item below:

- Three independent wide-format datasets selected from Discussion 5A
- Raw data files committed to GitHub (Step 3.1)
- Tidying pipeline documented and executable in `.Qmd` (Step 3.2)
- Analysis uses only the tidy dataset (Step 3.3)
- GitHub repository is public and all links resolve
- Published RPubs output is accessible without login